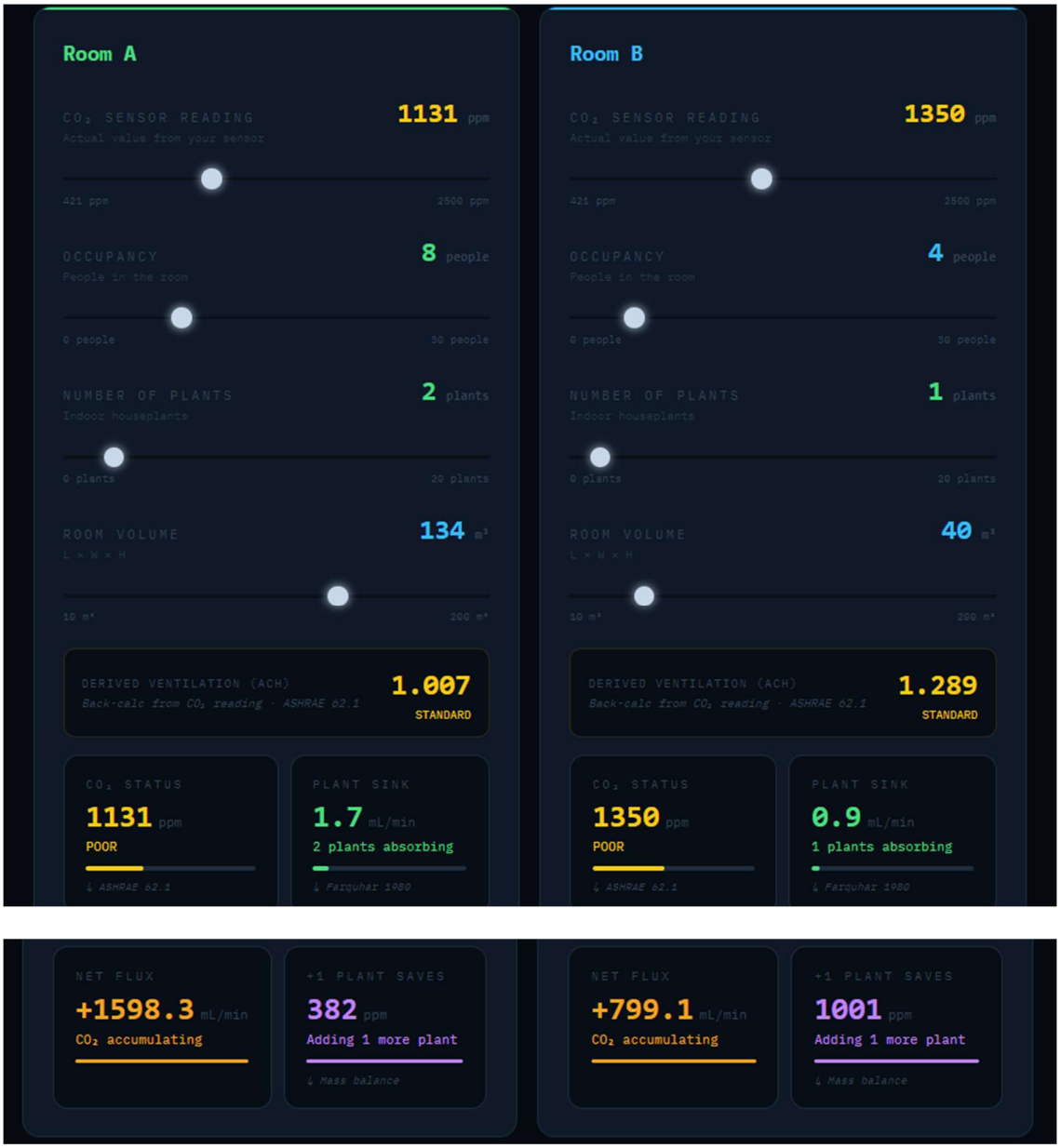
REAL TIME CO2 – FEATURES:

1. Chemical Analysis of enclosed space:
    1. How change in CO2 is related with number of plants in the room, size of room, and occupancy.



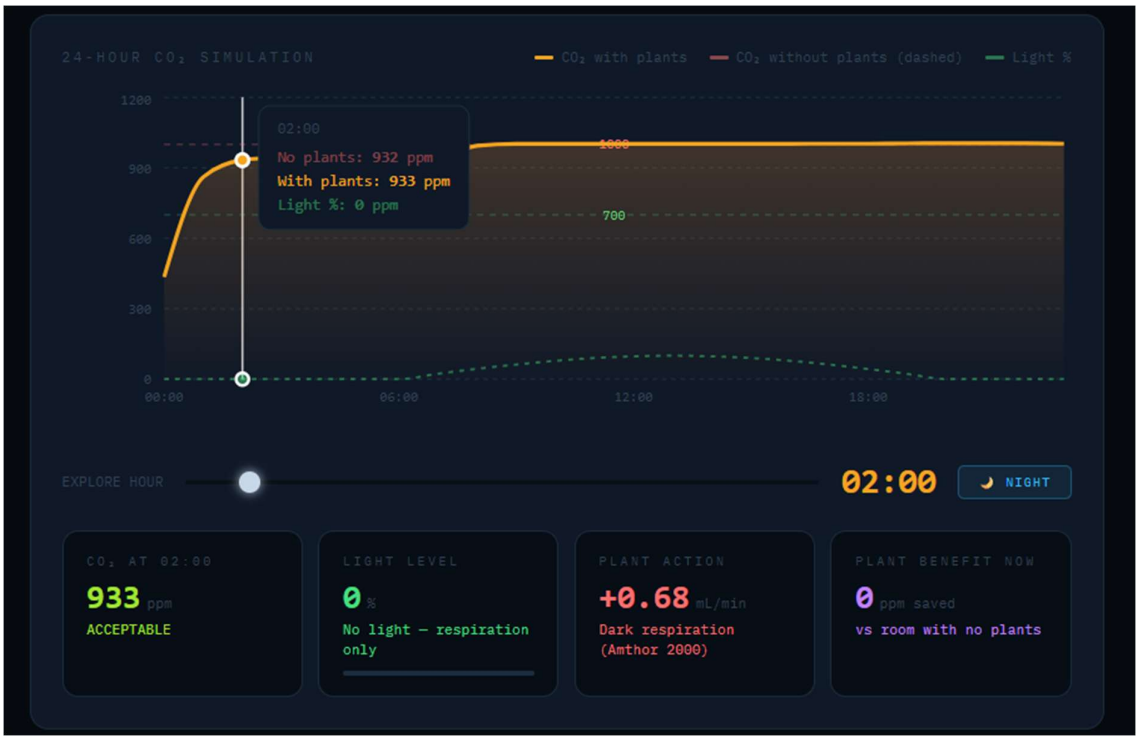
The Ventilation is calculated by ASHRAE 62.1 , it's the industrial standard for designing mechanical and natural ventilation system in commericials.
ACH = (N×200 − P×0.86) × 60 / (V × (CO_2_sensor − 420)) Back-calculated from sensor reading. Sources: ASHRAE 62.1, ISO 8996, Farquhar 1980.

Absorption by the plant is given by Farquhar 1980.

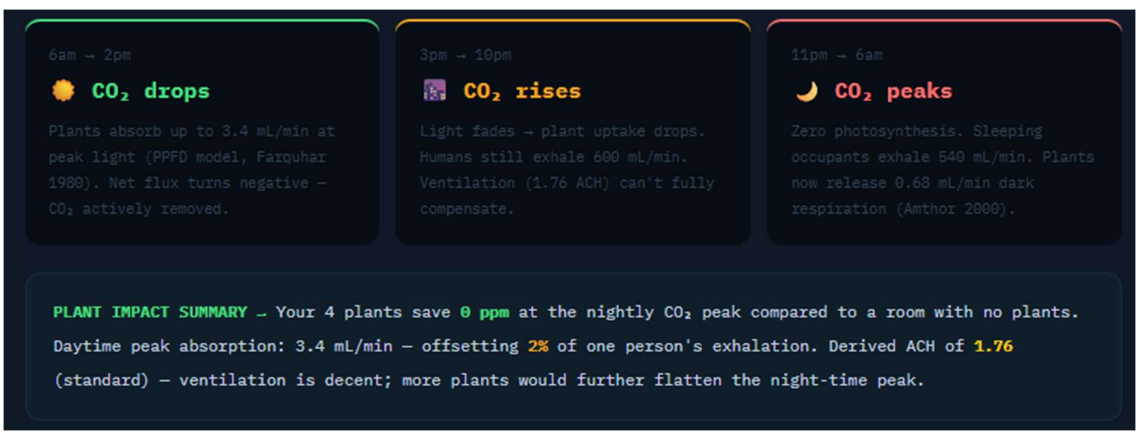
WHY ROOM A HAS LOWER CO_2 — Room A has 2 more plants, absorbing an extra 1.7 mL/min of CO_2 (Farquhar 1980). Each additional plant in Room A would reduce CO_2 by ~383 ppm; in Room B by ~1002 ppm. Both rooms are still accumulating CO_2 (net flux > 0) — plants help but ventilation still needs work. To reach 700 ppm: Room A needs ~1861 total plants, Room B ~931 total plants (at current occupancy).

Graph:

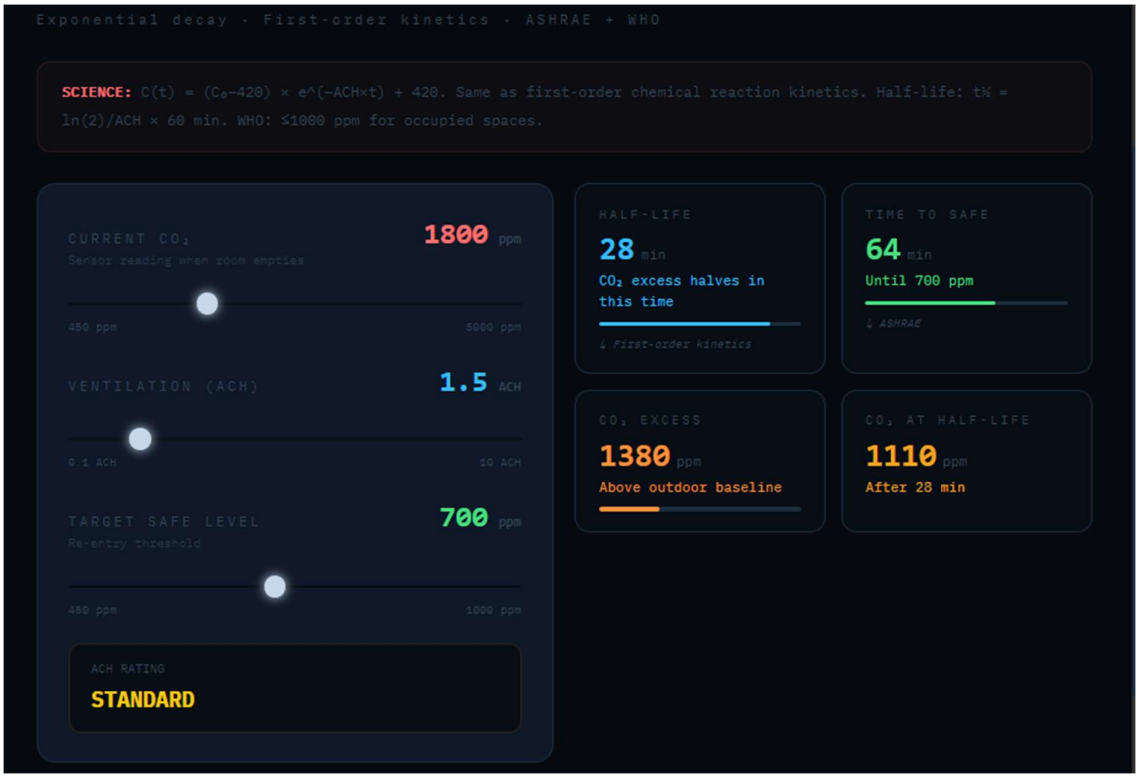
2. 24- Hour CO2 cylce (How CO2 concentration changes over day and night)



In this according to the timestamp, and co2 reading we can see how at night plants start releasing co2 at a rate of 0.68 ml/min (acc. To Amthor 200).



3. How fast will the CO2 will get cleared:



1. Features to the Users:

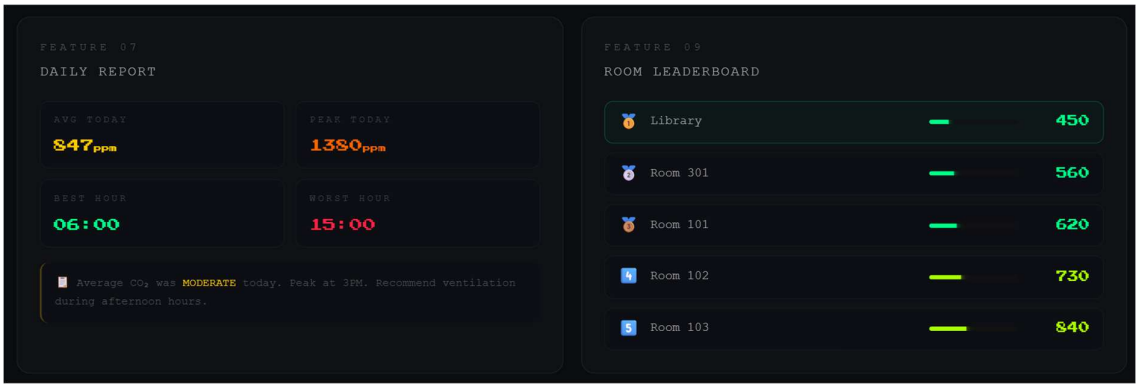
    1. Interactive Dashboard

Current Co2 level, along with a user credit system, where balance from the credit will be automatically deducted as co2 (ppm) increases. This credits resets everyday.

    2. Co2 Trend of your room:

3. Plant counting (manual), and Occupancy along with current co2 rising rate:

4. Room everyday report, and Leaderboard if there are multiple rooms:

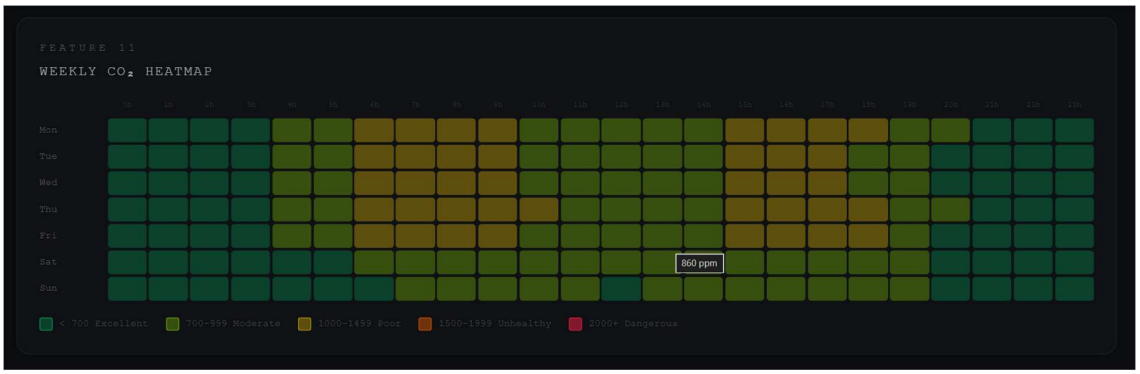


5. Set manual threshold:

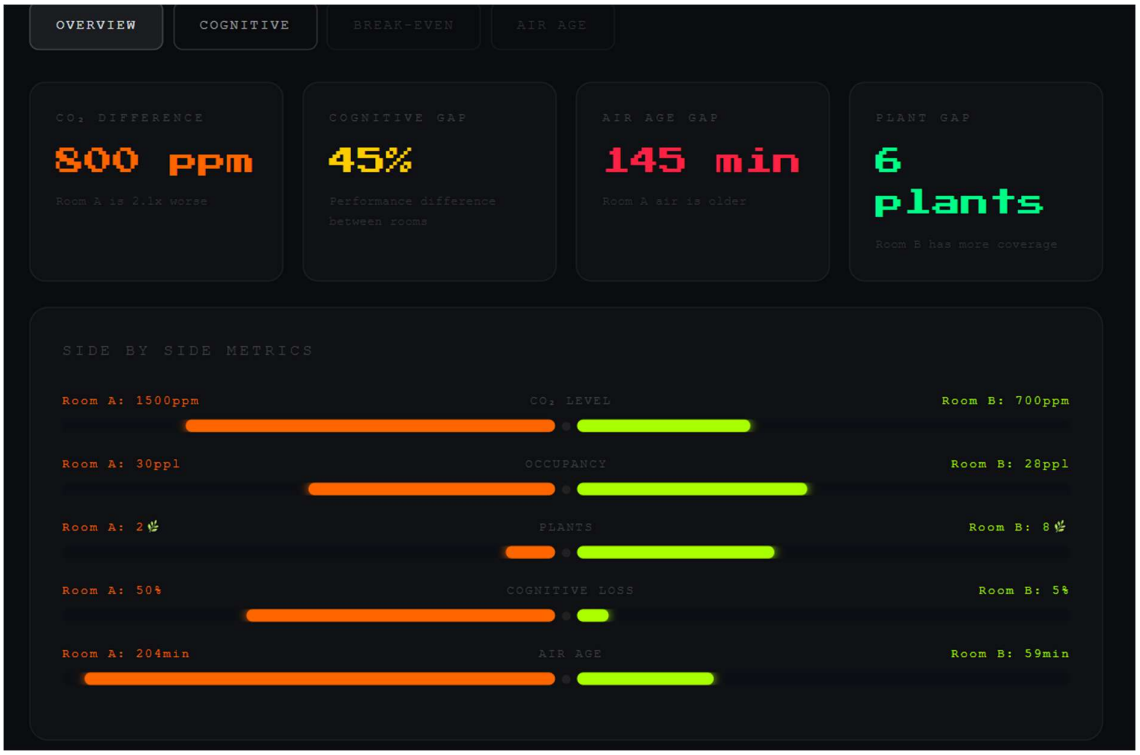
6. Prediction of the co2 (after 15min, after 45 min, after 60 min):

This prediction will be done by a machine learning model, so that we can make previous alert for better execution.

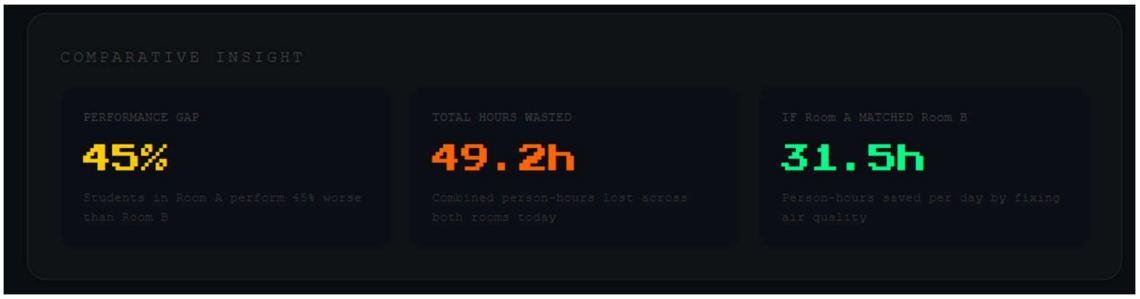
7. Weekly record/Heatmap:



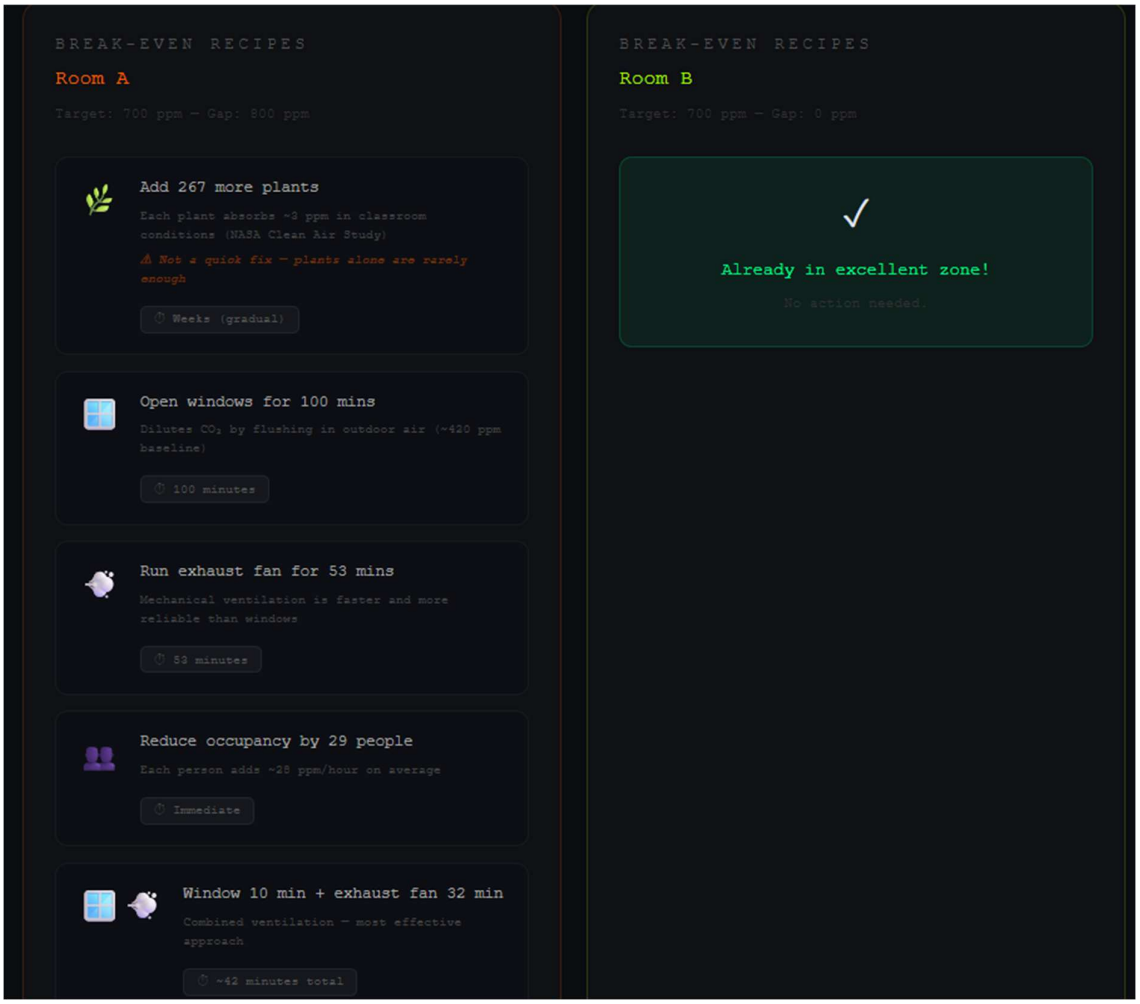
8. Comparison of 2 rooms together:
    1. Overview of 2 rooms:



2. Cognitive Analysis (Harvard T.H. Chan School of Public Health – 2015):



3. Things to do for lowering the Co2 :



4. MAA (Mean Air Age by HVAC Engineering). (Optional)